

IBTIDA

Comprehensive Technical Documentation

Architectural reference, domain services, native Kotlin/Swift modules, database schema, and complete screen-by-screen specifications for the Ibtida Islamic spiritual platform.

APPLICATION VERSION

1.0.0 (Expo SDK 57 / RN 0.86)

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DOCUMENT AUTHOR

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iOS, Android, Web

Ibtida (Ibadah v5) — Comprehensive Technical Handover & Architecture Specification

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1. Executive Summary & High-Level Architecture

Ibtida (derived from the Arabic word *Ibtidā'* meaning *Beginning / Genesis*) is an enterprise-grade Islamic spiritual companion, worship tracking, and digital mindfulness application. Ported with pixel-level fidelity from the *Ibadah* v5 design system specification, Ibtida combines classical Islamic tradition with modern mobile engineering paradigms.

Core Capabilities

1. **Real-time Astronomical Calculations:** High-precision prayer times calculated via the `adhan` astronomical engine, supporting multiple international conventions (MWL, ISNA, Umm Al-Qura, Karachi, Egypt, etc.), madhab-specific Asr timings, and dynamic spherical trigonometry for Kaaba Qibla bearing.
2. **"Ibadah Lock" (App-Blocking & Digital Fasting):** Enforces distraction-free worship through a custom native bridge (`modules/expo-ibadah-native`). On Android, an `AccessibilityService` halts distracting social/gaming apps during prayer or focus sessions; on iOS, Apple's `ScreenTime` (`FamilyControls` & `ManagedSettings`) framework provides hardware-level application shielding.
3. **Two-Stage Wake-Verified Prayer Alarm:** Combats sleep inertia and alarm snoozing by demanding physical proof of awakening: first scanning a QR code at the user's Wudu station, followed by a second verification QR scan on the prayer mat.
4. **Digital Tasbeeh & Verified Dhikr Corpus:** An interactive counter equipped with physics-based pearl bead rendering, haptic feedback profiles, session timers, and a curated library of authentic Quranic and Prophetic supplications across 8 life categories (with strict content governance forbidding unverified text).
5. **Community Circles & Collective Goals:** Social accountability structures allowing users to create private or invite-only circles, share dhikr milestones, and contribute personal counts toward massive collective targets (e.g., 1 Million Salawat).
6. **Kalimat Semantic Vector Search:** Neural search over the Holy Quran, Sahih/Sunan Hadith literature (graded Sahih/Hasan/Da'if), and Azkar corpus.



2. Technology Stack & Dependency Inventory

Core Frameworks

- **Runtime Environment:** React Native `0.86.3` powered by React `19.2.3`
- **Application Platform:** Expo SDK `~57.0.18` with custom native code (`expo-dev-client`)
- **Language:** TypeScript `~6.0.3` (strict type-checking enabled)
- **Navigation Engine:** `@react-navigation/native` `^7.3.18` , `@react-navigation/native-stack` `^7.18.10` , `@react-navigation/bottom-tabs` `^7.18.18`

Native Hardware, Sensor & System Libraries

- `expo-camera` (`~57.0.4`): Powers real-time camera viewfinder and barcode/QR scanning for wake alarms.
- `expo-location` (`~57.0.14`): High-accuracy device GPS acquisition and reverse geocoding for prayer calculations.
- `expo-notifications` (`~57.0.15`): Time-sensitive local notifications, Android notification channels, interruption levels, and deep-link tap routing.
- `expo-haptics` (`^57.0.2`): Tactile haptic feedback engine (selection, light impact, medium impact, and custom milestone patterns).
- `expo-linear-gradient` (`^57.0.1`): Multi-stop sky gradient rendering across diurnal prayer phases.

- `expo-blur` (`^57.0.2`): Glassmorphism and frosted backdrop filters for modals and HUD controls.
- `expo-crypto` (`~57.0.2`): Cryptographically secure random UUID and token generation for QR tokens and invite codes.
- `expo-auth-session` & `expo-web-browser` : Secure OAuth 2.0 PKCE redirection and system browser authentication.
- `react-native-qrcode-svg` (`^6.3.21`): Vector-based SVG generation of verifiable QR codes for prayer mats and wudu stations.
- `react-native-svg` (`15.15.4`): Custom vector icon system, SVG circular progress rings, and pearl bead strands.

Specialized Domain Engines

- `adhan` (`4.4.4`): Astronomical prayer time calculation library implementing standard solar declination algorithms.
- `@supabase/supabase-js` (`^2.112.4`): PostgreSQL database client with JWT auto-refresh, Auth, and Storage.
- `Kalimat API v2` : Cloud-based semantic vector search engine for Quranic verses, Hadith chapters/narrations, and Azkar.

Typography

- `@expo-google-fonts/scheherazade-new` (`^0.4.3`): Traditional Arabic typeface (`ScheherazadeNew_400Regular` , `_500Medium` , `_600SemiBold` , `_700Bold`) supporting complex Arabic ligatures and tashkeel diacritics.
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3. Repository Directory Map

```
Ibtida/
├── App.tsx # Application root, font hydration, auth gate & deep-link routing
├── app.json # Expo configuration, permissions, plugins & scheme definitions
├── eas.json # EAS Build profiles (development, preview, production)
├── package.json # Dependencies, engines and script definitions
├── tsconfig.json # TypeScript compiler configurations
├── assets/ # Static visual assets (Hadith cards, logos, splashes)
├── docs/ # Architecture and compliance documents
│   ├── HANDOVER_TECHNICAL_DOCUMENTATION.md # THIS COMPREHENSIVE DOCUMENT
│   ├── ios-family-controls-entitlement.md # Apple FamilyControls approval guide
│   └── native-features-testing.md # Manual testing protocols for native modules
├── modules/
│   └── expo-ibadah-native/ # Local native Expo module (Kotlin & Swift)
│       ├── expo-module.config.json # Autolinking specification
│       ├── index.ts # TypeScript native contract & JS wrappers
│       ├── android/ # Android AccessibilityService & AlarmManager
│       └── ios/ # iOS FamilyControls & ManagedSettings Shield
├── plugins/
│   └── withIbadahAndroidManifest.js # Expo config plugin for Android lock-screen flags
├── src/
│   ├── components/ # 19 Shared UI components (atomic & composite)
│   ├── lib/ # Foundation clients (Supabase, Adhan, Kalimat)
│   ├── navigation/ # Navigation types, custom tab bar, and imperative nav helpers
│   ├── screens/ # 22+ feature screens organized by domain folder
│   │   ├── onboarding/ # WelcomeScreen, IntentionsScreen
│   │   ├── home/ # HomeScreen (living sky world & daily insights)
│   │   ├── prayer/ # PrayerScreen, PrayerDetailScreen, LocationSettingsSheet
│   │   ├── adhkar/ # AdhkarScreen, Session, Tasbeeh, Goals, Progress
│   │   ├── quran/ # QuranScreen, QuranReaderScreen
│   │   ├── focus/ # FocusSetupScreen, FocusActiveScreen, EmergencyHistory
│   │   ├── community/ # CommunityScreen, Circles, Detail, New, Goals
│   │   ├── profile/ # ProfileScreen, PrivacyScreen
│   │   └── shared/ # Search, Notifications, WakeAlarmSettings, QR Scanner
│   ├── services/ # 12 Domain service modules wrapping Supabase CRUD
│   ├── state/ # AppState, AuthContext, static content libraries
│   └── theme/ # Design tokens, color palettes, and SVG icons
├── supabase/
│   ├── README.md # Migration execution instructions
│   └── migrations/ # 16 Sequential SQL migrations with RLS policies
```

4. Core Architectural Subsystems

4.1. Authentication, Session Lifecycle & Onboarding Gate

The authentication layer resides in `src/state/AuthContext.tsx` and encapsulates all interaction with Supabase Auth:

1. Authentication Providers Supported:

- Standard Email & Password (`signUpWithEmail` , `signInWithEmail`)
- Passwordless Magic Link (`signInWithMagicLink`)
- Google OAuth (`signInWithGoogle`) via `expo-auth-session` and `expo-web-browser`

2. Deep-Link Callback Flow:

- Standalone scheme: `ibtida://auth/callback`
- Handles both query-based PKCE exchange (`?code=...`) and hash-fragment session payloads (`#access_token=...`).
- Automatically invokes `supabase.auth.setSession` or `supabase.auth.exchangeCodeForSession` .

3. Background Token Refresh Management:

- Mobile operating systems do not fire browser window focus events. An active listener on React Native's `AppState` ('active' vs 'background') triggers `supabase.auth.startAutoRefresh()` when foregrounded and `supabase.auth.stopAutoRefresh()` when backgrounded to conserve battery and eliminate token desynchronization.

4. Onboarding Gate Logic in `App.tsx` :

- An asynchronous check evaluates `isOnboardingComplete(userId)` stored in `AsyncStorage` (`ibtida.onboarded.<userId>`).
- Navigation tree evaluation:
 - Unauthenticated $\searrow$ `WelcomeScreen`
 - Authenticated but First Launch $\searrow$ `IntentionsScreen`
 - Authenticated & Onboarded $\searrow$ `TabNavigator` (`Tabs`)

4.2. Astronomical Prayer-Time Engine & Qibla Geometry

Located in `src/lib/prayerTimes.ts` and backed by `src/services/prayerSettings.ts` :

1. Calculations:

- Leverages `adhan-js` (`CalculationMethod` , `Coordinates` , `Madhab` , `PrayerTimes`).
- Translates database parameters (`MuslimWorldLeague` , `Egyptian` , `Karachi` , `UmmAlQura` , `Dubai` , `Qatar` , `Kuwait` , `MoonsightingCommittee` , `Singapore` , `NorthAmerica` , `Tehran` , `Turkey`) directly into solar declination equations.
- Computes exact moments for: **Fajr**, **Sunrise**, **Dhuhr**, **Asr**, **Maghrib**, and **Isha**.

2. Asr Jurisprudence (Madhab):

- `Shafi` / standard: Shadow length equals object height ($1 \times$).
- `Hanafi` : Shadow length equals twice object height ($2 \times$).

3. Qibla Direction (Spherical Trigonometry):

- Kaaba coordinates: Latitude $(21.422487^\circ \text{ N})$, Longitude $(39.826206^\circ \text{ E})$.
- Great-circle forward azimuth formula:
$$\theta = \arctan2\left(\sin(\Delta\lambda)\cos(\phi_2), \cos(\phi_1)\sin(\phi_2) - \sin(\phi_1)\cos(\phi_2)\cos(\Delta\lambda)\right)$$
- Returns normalized heading degrees ($0^\circ - 360^\circ$) from true north.

4. Dynamic Countdown Window:

- `getPrayerCountdownWindow(settings, now)` automatically resolves current prayer, upcoming prayer, and exact milliseconds remaining, seamlessly bridging the midnight boundary (Isha to Fajr next day).

4.3. Motivational & Prayer Alert Notification Architecture

Located in `src/services/prayerNotifications.ts` :

1. The Daily Drift Problem:

- Islamic prayer times change by 1–2 minutes daily due to seasonal solar declination. Standard repeating alarms (e.g., `daily at 05:12`) drift into invalidity within days.

2. Rolling 7-Day Window Solution:

- Pre-computes exact dates and times for 5 prayers across a rolling 7-day window.
- Schedules one-shot notifications with explicit `DATE` triggers.
- Resynchronizes seamlessly on every app foreground transition (`AppState.addListener('change')`).

3. iOS Notification Ceiling Management:

- iOS enforces a hard limit of 64 pending local notifications per app.
- Allocation: 5 prayers $\times$ 7 days $(= 35)$ $+$ 3 dhikr chains $\times$ 7 days $(= 21)$ $(= 56)$ total notifications, safely below the 64-notification threshold while leaving headroom for Wake Alarms.

4. Rich Content Library:

- Rotates authentic hadith supplications and Quranic verses indexed by day of week (`Sun` through `Sat`).

5. Tap Routing:

- Notification payloads carry metadata: `{ kind: 'prayer-adhan' }` or `{ kind: 'dhikr-chain' }`, triggering immediate deep-linking to prayer or tasbeeh screens.

4.4. Ibadah Lock: Native App-Blocking & Digital Fasting Subsystem

Located in `src/services/ibadahLock.ts` and `modules/expo-ibadah-native`:

Ibadah Lock	
Android	iOS
- Mechanism: AccessibilityService	- Mechanism: ScreenTime / ManagedSettings
- Intercepts: TYPE_WINDOW_STATE_CHANGED	- Intercepts: Hardware shield overlay
- Enforcement: Immediate redirect to BlockedActivity	- Tokens: Opaque ApplicationToken
- Persistence: SharedPreferences	- Enforcement: OS-level app block
- Emergency: Dhikr challenge or manual override logged to Supabase	- Persistence: ManagedSettingsStore
	- Emergency: Stop session in Ibtida releases FamilyControls store

1. Android Engine (`IbadahBlockingAccessibilityService.kt`):

- Listens for `AccessibilityEvent.TYPE_WINDOW_STATE_CHANGED`.
- Compares the foreground package name against the blocked set stored in `IbadahBlockingPrefs`.
- If blocked, immediately launches `BlockedActivity` with flags `FLAG_ACTIVITY_NEW_TASK` | `FLAG_ACTIVITY_CLEAR_TOP`, presenting a serene interface reminding the user of their active worship session.
- Supports in-situ unblocking via Dhikr recitation challenge (`DhikrUnlockContent.kt`).

2. iOS Engine (`IbadahNativeModule.swift`):

- Interfaces directly with Apple's `FamilyControls` and `ManagedSettings` frameworks.

- Converts serialized base64 application tokens into `ApplicationToken` structures.
- Activates `ManagedSettingsStore().shield.applications = Set(tokens)`.
- Presents native iOS Shields without running background accessibility loops.
- *Requires Apple Family Controls Entitlement approval.*

3. Emergency Override Accountability:

- Acknowledging genuine emergencies, users can slide to unlock during an active lock session.
- Triggers `logEmergencyOverride(userId, sessionId, reason)` into Supabase `emergency_overrides`, creating a private, reflective accountability log visible in `EmergencyHistoryScreen`.

4.5. Two-Stage Wake-Verified Prayer Alarm (Wudu & Mat QR Verification)

Located in `src/services/wakeAlarmScheduling.ts` and `src/services/wakeAlarm.ts`:

1. The Problem of Sleep Inertia:

- Traditional alarms are dismissed reflexively in half-asleep states.

2. Two-Stage Physical Protocol:

- **Stage 1 (Wudu Station):** When the alarm rings, the user must walk to their washroom/wudu station and scan the Wudu QR tag. This cancels the immediate alarm and logs `LogWuduScan`.
- **10-Minute Mat Grace Window:** A 10-minute countdown is scheduled. If the user returns to bed and fails to reach the prayer mat, a secondary urgent alarm fires.
- **Stage 2 (Prayer Mat):** The user scans the unique QR code affixed to their prayer mat (`PrayerMatTagScreen`). This logs `logWakeVerification`, marks the prayer verified, and successfully concludes the awakening sequence.

3. Cryptographic Tag Integrity:

- QR tags encode unique random tokens stored in `prayer_alarm_settings` (`wudu_token`, `verification_token`).
- Tokens can be regenerated at any time from `PrayerMatTagScreen`.

4. Android Native Full-Screen Alarm:

- Utilizes `AlarmManager.setAlarmClock()` (immune to Android Doze mode).
- Rings through `WakeAlarmRingingService` and launches `WakeAlarmFullScreenActivity` over the keyguard/lockscreen.

4.6. Adhkar Engine, Digital Tasbeeh & Dhikr Bead Physics

Located in `src/screens/adhkar/TasbeehScreen.tsx` and `src/state/adhkarContent.ts`:

1. Bead Geometry & Orbital Mechanics:

- Renders 33 orbital beads on a dynamic SVG circular path with radial lighting gradients and 3D highlights on the active bead.
- Bead position calculation:
$$\begin{aligned} [x = \frac{\text{size}}{2} + r \cdot \cos\left(\frac{2\pi \cdot i}{33} - \frac{\pi}{2}\right), \quad y = \frac{\text{size}}{2} + r \cdot \sin\left(\frac{2\pi \cdot i}{33} - \frac{\pi}{2}\right)] \end{aligned}$$

2. Multi-Sensory Feedback Engine:

- Haptic triggers: Distinct haptic bursts for single count, interval milestones ($\times 10$), $\times 33$), and full goal completion.
- Undo protection: Long-press anywhere on the screen decrements count (-1) to correct accidental touches.

3. Per-Dhikr Atmospheric Visuals:

- Dynamic accent palette changes ring colors, particle glows, and background tints according to the active supplication (e.g., Emerald for *SubhanAllah*, Gold for *Alhamdulillah*, Amethyst for *AllahuAkbar*, Sapphire for *Astaghfirullah*, Crimson for *La ilaha illallah*).

4. Content Governance (§35 Policy):

- Absolute prohibition against AI-generated religious scripture. All 8 categories (*Morning, Evening, After Salah, Protection, Forgiveness, Gratitude, Before Sleep, Travel*) contain rigorously cited texts from *Sahih Al-Bukhari, Sahih Muslim, Sunan Abi Dawud, and Jami' at-Tirmidhi*.

4.7. Community Circles, Shared Goals & Collective Impact

Located in `src/services/community.ts` and `supabase/migrations/0006_community.sql`:

1. Circles & Membership:

- Circles have four privacy modes: `Invite only`, `Private`, `Friends`, `Public`.
- Distinct roles: `owner` (can manage settings, kick members, delete circle) and `member`.
- Unique 8-character cryptographic invite codes (`regenerateInviteCode`) shareable via standard native share sheets.

2. Community Goal Aggregation:

- Goals represent collective milestones (e.g., *1,000,000 Durood Sharif*).
- When a user performs Tasbeeh linked to a community goal, `updateGoalProgress` executes a dual write: updating their personal goal row in `adhkar_goals` and upserting their participant contribution in `community_goal_members`.
- The total circle progress is computed as $\sum \text{progress}$ across all participants.

4.8. Kalimat AI Semantic Search Integration

Located in `src/lib/kalimatApi.ts` and `src/screens/shared/SearchScreen.tsx`:

1. Neural Vector Search:

- Queries `api.kalimat.dev/api/v2` with `X-API-Key` authentication.
- Searches across three distinct corpora:
 - `quran`: Matches verses, surahs, and juz by semantic meaning in English or Arabic.
 - `sunnah`: Searches major Hadith compilations (*Bukhari, Muslim, Abu Dawood, Tirmidhi, Nasai, Ibn Majah*) with grade filtering (*Sahih, Hasan, Da'if*).
 - `azkar`: Supplications and daily invocations.

2. Client Performance & Debouncing:

- 350ms input debouncing prevents API flooding.
- Active `AbortController` terminates stale in-flight HTTP requests when the user continues typing.

4.9. Design System, Color Science & Dual-Theme Tokens

Located in `src/theme/tokens.ts` :

Token Category	Light Mode Value	Dark Mode (Moonly Deep Space)	Semantic Role
Surface Bg	<code>#FFFFFF</code>	<code>#0F1419</code>	Main screen canvas background
Surface Wash	<code>#F8F7F3</code>	<code>#141C2C</code>	Secondary card & container wash
Ink Primary	<code>#1B2430</code>	<code>#EFF3F0</code>	High-contrast body & header typography
Ink Muted	<code>#68716C</code>	<code>rgba(239, 243, 240, 0.38)</code>	Subtitles, footnotes & timestamps
Primary Blue	<code>#2A63B8</code>	<code>#3B7DDE</code>	Action buttons, active tabs & indicators
Success Green	<code>#5EAA78</code>	<code>#5EAA78</code>	Completed prayers, milestones & streaks
Gold / Warm	<code>#D4A86A</code>	<code>#D4A86A</code>	Beads, divine accents, special rings
Fajr	<code>#4A7FC1</code>	<code>#4A7FC1</code>	Dawn blue
Dhuhr	<code>#D9822E</code>	<code>#D9822E</code>	Solar midday amber
Asr	<code>#5EAA78</code>	<code>#5EAA78</code>	Afternoon emerald
Maghrib	<code>#C0563F</code>	<code>#C0563F</code>	Sunset crimson
Isha	<code>#2F4B6E</code>	<code>#2F4B6E</code>	Midnight navy

5. Exhaustive Screen-by-Screen Reference (All Screens)

5.1. Onboarding Domain

1. `WelcomeScreen`

- **File:** `src/screens/onboarding/WelcomeScreen.tsx`
- **Route:** `Welcome` (`RootStackParamList`)
- **Visual Design:** Hero geometric Islamic arch illustration, gold star accents, dual-mode auth container (Sign In vs Sign Up), floating ivory inputs.
- **Functionality:**
 - Email/password authentication with live validation.
 - One-tap Google OAuth modal trigger.
 - Passwordless Magic Link dispatcher with informative "Check your inbox" confirmation.
 - Seamless toggle between Sign In and Sign Up modes.

2. IntentionsScreen

- **File:** `src/screens/onboarding/IntentionsScreen.tsx`
 - **Route:** `Intentions` (`RootStackParamList`)
 - **Visual Design:** Multi-select spiritual intention checklist, interactive check pills, serene introduction copy.
 - **Functionality:**
 - Lets users choose spiritual priorities (e.g., *Guard the 5 daily prayers, Build a morning adhkar habit, Read Quran regularly, Digital fasting/focus*).
 - Persists onboarding completion status in `AsyncStorage` (`ibtida.onboarded.<userId> = '1'`).
 - Navigates directly to `Tabs` (`DashboardTab`).
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5.2. Dashboard / Home Domain

3. HomeScreen

- **File:** `src/screens/home/HomeScreen.tsx`
 - **Route:** `Home` (`HomeStackParamList`)
 - **Visual Design: Living World Hero Sky Scene:**
 - 5 dynamic sky gradients that cross-fade in real time based on active prayer phase (*Fajr, Dhuhr, Asr, Maghrib, Isha*).
 - Ambient breathing background orbs (9s and 14s periodic animation cycles).
 - Twinkling star field spread across the upper 60% of the screen.
 - Hijri calendar date display (e.g., *14 Ramadan 1447 AH*).
 - Horizontal timeline bar with pulsating indicator on current prayer.
 - Auto-scrolling Hadith wisdom carousel with pagination dots.
 - Community Impact Counter with animated easing numbers.
 - **Functionality:**
 - Automatically recalculates prayer countdown every second.
 - Direct quick-action cards: *Prayer Times, Daily Adhkar, Quran Reader, Focus Mode*.
 - Pull-down trigger or button to present the comprehensive **Daily Insight Bottom Sheet**.
-

5.3. Prayer Domain

4. PrayerScreen

- **File:** `src/screens/prayer/PrayerScreen.tsx`
- **Route:** `PrayersTab` (`TabParamList`)
- **Visual Design:** 7-day horizontal calendar date strip, large circular countdown ring with real-time seconds, five prayer cards displaying status (done, current, upcoming), expandable Qibla compass bearing card.
- **Functionality:**
 - Displays computed prayer times for the selected date.
 - One-tap prayer logging: Updates `prayer_logs` table in Supabase with optimistic UI updates.
 - Adhan toggle: Configures notification alerts per prayer (`adhan_settings`).

- Qibla compass card calculates direct bearing towards Mecca.
- Header actions: Direct access to `PrayerLocationSettingsSheet` and `WakeAlarmSettingsScreen`.

5. `PrayerDetailScreen`

- **File:** `src/screens/prayer/PrayerDetailScreen.tsx`
- **Route:** `PrayerDetail` (`{ prayerName: string }`)
- **Visual Design:** Half-sheet transparent modal, prayer status badge, countdown timer, Adhan sound selector, rak'ah breakdown (Fard, Sunnah, Nafil).
- **Functionality:**
 - Allows marking prayer as prayed on time or missed.
 - Toggles audio adhan reminder settings.
 - Direct link to `PrayerMatTagScreen` for wake verification setup.

6. `PrayerLocationSettingsSheet`

- **File:** `src/screens/prayer/PrayerLocationSettingsSheet.tsx`
- **Route:** Modal sheet rendered within `PrayerScreen`
- **Visual Design:** Form with location detection button, city/country labels, calculation method selector, and madhab picker.
- **Functionality:**
 - Requests GPS permissions via `expo-location` and fetches coordinates.
 - Performs reverse geocoding to resolve city/region.
 - Upserts settings to Supabase `prayer_calc_settings`.
 - Re-triggers prayer calculation and notification rescheduling across the entire app.

5.4. Adhkar & Goals Domain

7. `AdhkarScreen`

- **File:** `src/screens/adhkar/AdhkarScreen.tsx`
- **Route:** `Adhkar` (`AdhkarStackParamList`)
- **Visual Design:** Mode selector (*Categories* vs *Personal Goals*), 8 rich category cards with Arabic calligraphy labels, estimated completion time, and progress bars.
- **Functionality:**
 - Tapping a category launches `AdhkarSessionScreen`.
 - Switching to Goals displays active user goals with streak counts and completion rings.
 - Header button allows creating new goals (`GoalNewScreen`).

8. `AdhkarSessionScreen`

- **File:** `src/screens/adhkar/AdhkarSessionScreen.tsx`
- **Route:** `AdhkarSession` (`{ category: string }`)
- **Visual Design:** Full-screen immersive reader, large Arabic script (`Scheherazade New`), transliteration, English translation, verified Hadith source citation, bottom progress bar, and tap counter.
- **Functionality:**

- Advances through category supplications one by one upon reaching the prescribed count (e.g., \ (1\times\), \ (3\times\), \ (7\times\), \ (33\times\)).
- Full-screen tap interaction with haptic feedback.
- Auto-completes and returns to dashboard with congratulatory toast upon finishing the final dhikr.

9. TasbeehScreen

- **File:** `src/screens/adhkar/TasbeehScreen.tsx`
- **Route:** `Tasbeeh2` (`AdhkarStackParamList`)
- **Visual Design:** Cosmic dark background with twinkling stars, central SVG progress ring, 33 3D-shaded pearl beads, live elapsed session timer, Niyah/hadith benefit card, frosted-glass dhikr selection carousel.
- **Functionality:**
 - Tap anywhere to count with custom haptic curves.
 - Long press to undo count.
 - Vibration toggle in header.
 - Saves completed counts to Supabase `tasbeeh_sessions` .

10. GoalsScreen

- **File:** `src/screens/adhkar/GoalsScreen.tsx`
- **Route:** `Goals` (`AdhkarStackParamList`)
- **Visual Design:** Active personal goals list, 7-day streak dot rows, circular progress rings, reminder times.
- **Functionality:**
 - Tapping a goal opens `GoalTasbeehScreen` to immediately practice dhikr toward that target.
 - Action button to add a new goal (`GoalNewScreen`).

11. GoalNewScreen

- **File:** `src/screens/adhkar/GoalNewScreen.tsx`
- **Route:** `GoalNew` (`RootStackParamList`)
- **Visual Design:** Segmented control (*Community Linked* vs *Personal Goal*), preset dhikr selector, custom target counter input, frequency picker.
- **Functionality:**
 - Community mode fetches active collective goals from `community_goals` so users can link their personal effort to an Ummah-wide initiative.
 - Inserts new row to `adhkar_goals` via `createGoal` .

12. GoalScheduleScreen

- **File:** `src/screens/adhkar/GoalScheduleScreen.tsx`
- **Route:** `GoalSchedule` (`RootStackParamList`)
- **Visual Design:** Time picker, day-of-week checkboxes, notification tone preview.
- **Functionality:** Configures daily reminder schedules for individual dhikr goals.

13. GoalTasbeehScreen

- **File:** `src/screens/adhkar/GoalTasbeehScreen.tsx`
- **Route:** `GoalTasbeeh` (`{ goalId, title, target, progress, communityGoalId }`)

- **Visual Design:** Goal-focused Tasbeeh counter showing real-time progress toward target, remaining counts, and community contribution badge.
- **Functionality:**
 - Dual-syncs counts to both personal `adhkar_goals` and collective `community_goal_members`.
 - Automatically transitions to `GoalCompleteScreen` when the target is reached.

14. `GoalCompleteScreen`

- **File:** `src/screens/adhkar/GoalCompleteScreen.tsx`
- **Route:** `GoalComplete` (`RootStackParamList`)
- **Visual Design:** Celebration illustration, golden trophy badge, summary of recitations completed, streak milestone announcement, share button.
- **Functionality:** Marks goal as completed in Supabase (`completed_at = now()`) and offers return to Home or Goals list.

15. `ProgressScreen`

- **File:** `src/screens/adhkar/ProgressScreen.tsx`
- **Route:** `Progress2` (`AdhkarStackParamList`)
- **Visual Design:** Range control (*Today, Week, Month, Year*), 4-metric statistics grid (*Prayers logged, Dhikr counted, Quran pages, Adhkar sessions*), bar chart of daily activity, 84-day contribution heat map grid.
- **Functionality:** Aggregates historic worship data across all domains to visualize consistency and spiritual growth.

5.5. Quran Domain

16. `QuranScreen`

- **File:** `src/screens/quran/QuranScreen.tsx`
- **Route:** `Quran` (`QuranStackParamList`)
- **Visual Design:** Three-tab segmented control (*Surahs, Juz, History*), search bar, Surah cards showing revelation order (Makkah/Madinah) and Ayah counts, Juz progress indicators.
- **Functionality:**
 - Tapping any Surah or Juz opens `QuranReaderScreen`.
 - Tracks reading history and saved bookmarks.

17. `QuranReaderScreen`

- **File:** `src/screens/quran/QuranReaderScreen.tsx`
- **Route:** `QuranReader` (`RootStackParamList`)
- **Visual Design:** Distraction-free Mushaf reading view, large Arabic font, ayah-by-ayah cards, bookmark toggle in navigation bar, reading settings sheet (font size, translation toggles).
- **Functionality:**
 - Strict compliance with Content Governance §35: Shows licensed translation loading state.
 - Persists user bookmarks to Supabase `quran_bookmarks`.

5.6. Focus & Ibadah Lock Domain

18. FocusSetupScreen

- **File:** `src/screens/focus/FocusSetupScreen.tsx`
- **Route:** `FocusSetup` (`RootStackParamList`)
- **Visual Design:** Duration selector (*Until goal completed, 15m, 30m, 1h*), active goal picker, blocked apps list with platform indicator (iOS ScreenTime / Android Accessibility), app picker trigger button.
- **Functionality:**
 - Checks native blocking support (`isAppBlockingSupported`).
 - Launches native app picker (`pickAppsToBlock`).
 - Starts locked session via `startGoalLockedSession` and navigates to `FocusActiveScreen` .

19. FocusActiveScreen

- **File:** `src/screens/focus/FocusActiveScreen.tsx`
- **Route:** `FocusActive` ({ `goalId?: string`, `target?: number` })
- **Visual Design:** High-focus dark HUD, circular dhikr bead ring, countdown timer, target progress, "Slide to Emergency Unlock" bottom slider with confirmation modal.
- **Functionality:**
 - Enforces active app blocking via `startBlocking` .
 - Tapping counts toward linked goal.
 - Reaching target automatically releases app blocks (`stopBlocking`) and completes session.
 - Sliding emergency unlock prompts confirmation and logs reason to `emergency_overrides` .

20. EmergencyHistoryScreen

- **File:** `src/screens/focus/EmergencyHistoryScreen.tsx`
- **Route:** `EmergencyHistory` (`RootStackParamList`)
- **Visual Design:** Private, reflective audit log of past emergency overrides with timestamps and stated reasons.
- **Functionality:** Fetches `emergency_overrides` from Supabase for personal reflection and mindfulness accountability.

5.7. Community & Circles Domain

21. CommunityScreen

- **File:** `src/screens/community/CommunityScreen.tsx`
- **Route:** `Community` (`CommunityStackParamList`)
- **Visual Design:** Tab strip (*Global Goals, My Circles*), total Ummah impact counter, active community goal cards with progress bars and participant counts, circle roster previews.
- **Functionality:**
 - Join global community goals.
 - Access circle management (`CirclesScreen`).
 - Create new circle (`CircleNewScreen`).

22. CirclesScreen

- **File:** `src/screens/community/CirclesScreen.tsx`
- **Route:** `Circles2` (`CommunityStackParamList`)
- **Visual Design:** List of joined circles, privacy status badges (`Private` , `Invite only` , `Friends`), member count badges, join circle via invite code input.
- **Functionality:**
 - Join circle by pasting 8-character invite code (`joinCircleByCode`).
 - Tap circle to open `CircleDetailScreen` .

23. CircleDetailScreen

- **File:** `src/screens/community/CircleDetailScreen.tsx`
- **Route:** `CircleDetail` ({ `circleId: string` })
- **Visual Design:** Circle header with privacy badge, shareable invite code card with copy/share buttons, member avatar stack and list with role badges (`Owner` , `Member`), circle-specific shared goals list with "+ Add Goal" button.
- **Functionality:**
 - Share invite code via native share dialog (`Share.share`).
 - Regenerate invite code (owner only).
 - Add circle-specific collective goal.
 - Contribute dhikr directly to circle goals.
 - Manage members (kick member, leave circle, delete circle).

24. CircleNewScreen

- **File:** `src/screens/community/CircleNewScreen.tsx`
- **Route:** `CircleNew` (`RootStackParamList`)
- **Visual Design:** Circle creation form with name input, privacy mode cards with explanatory subtext.
- **Functionality:** Creates circle in Supabase `community_circles` and sets creator as `owner` member.

25. CommunityGoalScreen

- **File:** `src/screens/community/CommunityGoalScreen.tsx`
- **Route:** `CommunityGoal2` ({ `id: number` })
- **Visual Design:** Deep-dive view of a specific community goal, overall progress percentage, collective recitation total, personal contribution count, recent contributor roster.
- **Functionality:** One-tap participation and direct linkage to personal Tasbeeh.

5.8. Profile & Privacy Domain

26. ProfileScreen

- **File:** `src/screens/profile/ProfileScreen.tsx`
- **Route:** `Profile` (`ProfileStackParamList`)
- **Visual Design:** Dark-theme profile card with user initials avatar, display name, 7-day prayer consistency ring, spiritual statistics grid (active goals, bookmarks, circles, streaks), calculation method summary badge,

navigation menu.

- **Functionality:**
 - Displays user profile and stats.
 - Navigation to `PrivacyScreen` , `WakeAlarmSettingsScreen` , `EmergencyHistoryScreen` .
 - Secure sign-out confirmation dialog with session teardown.

27. `PrivacyScreen`

- **File:** `src/screens/profile/PrivacyScreen.tsx`
 - **Route:** `Privacy2` (`ProfileStackParamList`)
 - **Visual Design:** Privacy controls menu (*Profile visibility, Activity visibility, Community participation, Goal visibility, Location, Analytics*).
 - **Functionality:** Reads and writes privacy preferences to Supabase `privacy_settings` .
-

5.9. Shared, Search & Native Verification Modals

28. `SearchScreen`

- **File:** `src/screens/shared/SearchScreen.tsx`
- **Route:** `Search` (`RootStackParamList`)
- **Visual Design:** Search input with clear button, content filter pills (*Quran, Hadith, Azkar*), quick suggestion tags, structured result cards with distinct color-coded iconography and metadata tags.
- **Functionality:** Connects to Kalimat API v2 with debouncing and request cancellation to display semantic matches across Islamic scripture.

29. `NotificationsScreen`

- **File:** `src/screens/shared/NotificationsScreen.tsx`
- **Route:** `Notifications` (`RootStackParamList`)
- **Visual Design:** Granular notification toggle list across categories (*Prayer, Adhkar, Goals, Quran, Focus, Community*).
- **Functionality:** Configures preferences in Supabase `notification_settings` .

30. `WakeAlarmSettingsScreen`

- **File:** `src/screens/shared/WakeAlarmSettingsScreen.tsx`
- **Route:** `WakeAlarmSettings` (`RootStackParamList`)
- **Visual Design:** Per-prayer toggle cards for wake verification, printable QR tag button, honest disclosure notice regarding mobile notification vs alarm clock limits.
- **Functionality:** Toggles wake-verification requirement per prayer in `prayer_alarm_settings` and schedules rolling notification alerts.

31. `PrayerMatTagScreen`

- **File:** `src/screens/shared/PrayerMatTagScreen.tsx`
- **Route:** `PrayerMatTag` (`{ prayerName: string }`)
- **Visual Design:** Printable vector QR code generated via `react-native-qrcode-svg` , instructions for affixing to prayer mat or wudu area, "Regenerate Tag" button.

- **Functionality:** Displays the cryptographic token stored in `prayer_alarm_settings` or generates a fresh token if compromised.

32. `WakeAlarmScanScreen` & `WakeScanScreen`

- **Files:** `src/screens/shared/WakeAlarmScanScreen.tsx`, `WakeScanScreen.tsx`
- **Route:** `WakeScan` ({ prayerName, alarmDate, stage })
- **Visual Design:** Full-screen camera viewfinder (`expo-camera`), targeting reticle overlay, torch toggle, stage indicator (*Stage 1: Wudu Station* \(\rightarrow\) *Stage 2: Prayer Mat*).
- **Functionality:**
 - Scans QR payload and compares against stored tokens in `prayer_alarm_settings`.
 - Stage 1 match logs `logWuduScan` and schedules 10-minute mat grace window.
 - Stage 2 match logs `logWakeVerification`, cancels pending follow-ups, halts Android ringing service, and dismisses modal.

33. `DailyInsightSheet`

- **File:** `src/components/DailyInsightSheet.tsx`
- **Component:** 880-line interactive bottom sheet modal launched from HomeScreen.
- **Visual Design:** Full spiritual dashboard containing:
 - Personal greeting & streak badge
 - Earned milestones (*3-Day Spark, 7-Day Warrior, 2-Week Steadfast, 30-Day Devoted, 60-Day Khushoo, Centennial Mujahid*)
 - Today's prayer completion ring & checklist
 - Active goals progress bars
 - Dawah network circle cards with instant invite code copying
 - Community collective dhikr contributions
- **Functionality:** Aggregated single-query fetch via `src/services/insights.ts` for instant rendering with skeleton states.

34. `ErrorStateScreen`

- **File:** `src/screens/shared/ErrorStateScreen.tsx`
- **Route:** `ErrorState` (`RootStackParamList`)
- **Visual Design:** Calm offline/error illustration, message copy, "Retry Connection" primary button.

6. Reusable UI Component Library

All components live in `src/components/` and adhere to strict design tokens:

Component	File	Description & Props
<code>ProgressRing</code>	<code>ProgressRing.tsx</code>	SVG circular progress indicator with animated stroke-dashoffset, custom stroke widths, and background tracks.
<code>SegmentedControl</code>	<code>SegmentedControl.tsx</code>	Animated tab strip supporting arbitrary options with slide-pill animation and haptic feedback.
<code>PressableScale</code>	<code>PressableScale.tsx</code>	Micro-interaction wrapper that scales down slightly ($(0.96 \times - 0.98 \times)$) on touch press with haptic buzz.
<code>StreakDotRow</code>	<code>StreakDotRow.tsx</code>	7-day dot indicator for prayer or goal habits with hit, missed, and pending states.
<code>BottomSheetModal</code>	<code>BottomSheetModal.tsx</code>	Accessible slide-up modal with backdrop overlay, drag handle, and dismiss gesture.
<code>ConfirmSheet</code>	<code>ConfirmSheet.tsx</code>	Reusable confirmation bottom sheet for destructive actions (sign out, delete circle, emergency unlock).
<code>DailyInsightSheet</code>	<code>DailyInsightSheet.tsx</code>	880-line composite spiritual overview modal.
<code>Card</code>	<code>Card.tsx</code>	Standard elevated surface container with token-matched borders and shadows.
<code>AvatarStack</code>	<code>AvatarStack.tsx</code>	Overlapping circular avatar cluster showing initials and member counts.
<code>BarChart</code>	<code>BarChart.tsx</code>	Lightweight vector bar chart for daily and weekly progress visualization.
<code>StatusPill</code>	<code>StatusPill.tsx</code>	Small rounded badge displaying semantic states (<i>Done</i> , <i>Current</i> , <i>Upcoming</i> , <i>Missed</i>).
<code>IconBadgeRow</code>	<code>IconBadgeRow.tsx</code>	Standard settings item with left icon badge, title, subtitle, and right accessory/chevron.
<code>Toggle</code>	<code>Toggle.tsx</code>	Custom animated boolean switch matching design system colors.
<code>Toast</code>	<code>Toast.tsx</code>	Floating pill notification banner that rises from screen bottom with auto-dismiss.
<code>Skeleton</code>	<code>Skeleton.tsx</code>	Shimmering loading placeholders (<code>SkeletonBlock</code> , <code>RowSkeleton</code> , <code>HomeSkeleton</code>).
<code>EmptyState</code>	<code>EmptyState.tsx</code>	Centered illustration, title, and action button for zero-data views.
<code>PrimaryButton</code>	<code>PrimaryButton.tsx</code>	Filled high-emphasis CTA button with press scale and loading spinner.
<code>SecondaryButton</code>	<code>SecondaryButton.tsx</code>	Outlined or ghost medium-emphasis button.
<code>ScreenFade</code>	<code>ScreenFade.tsx</code>	Page-level entrance animations (<code>ScreenFade</code> , <code>RiseIn</code>).

7. Supabase Database Schema, Migrations & RLS Policies

The Supabase PostgreSQL backend is defined across 16 migrations in [supabase/migrations/](#). Every table has **Row Level Security (RLS)** enabled.

Complete Database Schema Catalog

```
erDiagram
    profiles ||--o{ prayer_logs : "logs"
    profiles ||--o{ adhan_settings : "configures"
    profiles ||--o{ prayer_calc_settings : "has"
    profiles ||--o{ adhkar_goals : "creates"
    profiles ||--o{ tasbeeh_sessions : "records"
    profiles ||--o{ quran_bookmarks : "saves"
    profiles ||--o{ circle_members : "belongs to"
    profiles ||--o{ focus_sessions : "runs"
    profiles ||--o{ blocked_apps : "blocks"
    profiles ||--o{ emergency_overrides : "records"
    profiles ||--o{ prayer_alarm_settings : "sets"
    profiles ||--o{ wake_verifications : "proves"
    community_circles ||--o{ circle_members : "contains"
    community_circles ||--o{ community_goals : "scopes"
    community_goals ||--o{ community_goal_members : "tracks"
    adhkar_goals ||--o{ community_goals : "links to"
```

1. `profiles` (`0001_profiles.sql`)

- Primary key `id` UUID references `auth.users` on delete cascade .
- Columns: `display_name` text , `avatar_url` text , `updated_at` timestampz .
- Auto-created on signup via `handle_new_user` PostgreSQL trigger.
- **RLS:** Owner-only (`auth.uid() = id`).

2. `prayer_logs` (`0002_prayers.sql`)

- Composite key (`user_id`, `log_date`, `prayer_name`) .
- Columns: `prayer_name` prayer_name enum , `log_date` date , `done` boolean , `logged_at` timestampz .
- **RLS:** Owner-only (`auth.uid() = user_id`).

3. `adhan_settings` (`0002_prayers.sql`)

- Composite key (`user_id`, `prayer_name`) .
- Columns: `enabled` boolean default true .
- **RLS:** Owner-only (`auth.uid() = user_id`).

4. `adhkar_goals` (`0003_adhkar.sql` , `0012_adhkar_community_goal_link.sql`)

- Primary key `id` UUID default `gen_random_uuid()` .
- Columns: `user_id` UUID , `title` text , `target` int , `frequency` int , `range` int , `progress` int default 0 , `completed_at` timestampz , `community_goal_id` UUID references `community_goals` .
- **RLS:** Owner-only.

5. `tasbeeh_sessions` (`0003_adhkar.sql`)

- Primary key `id` UUID .

- Columns: `user_id UUID` , `dhikr_key text` , `count int` , `target int` , `duration_seconds int` , `created_at timestampz` .
 - RLS:** Owner-only.
6. `privacy_settings` & `notification_settings` (`0004_settings.sql`)
- Keyed by `(user_id, key text)` .
 - Columns: `value text` (privacy) or `enabled boolean` (notifications).
 - RLS:** Owner-only.
7. `quran_bookmarks` & `quran_reader_settings` (`0005_quran.sql`)
- Bookmarks: `surah_number int` , `ayah_number int` , `note text` .
 - Reader Settings: `font_size text` , `translation_id text` , `arabic_visible boolean` .
 - RLS:** Owner-only.
8. `community_circles` & `circle_members` (`0006` , `0008` , `0013` , `0014` , `0015`)
- `community_circles` : `id UUID` , `name text` , `privacy circle_privacy enum` , `created_by UUID` , `invite_code text unique` .
 - `circle_members` : `circle_id UUID` , `user_id UUID` , `role circle_role enum ('owner', 'member')` , `joined_at timestampz` .
 - RLS Policies:**
 - Fixed in `0013_fix_circle_qls_recursion.sql` to avoid infinite recursion by using direct subqueries.
 - A user can view circles where `id in (select circle_id from circle_members where user_id = auth.uid())` OR where `privacy = 'Public'` .
 - Members can view fellow member records within their circles.
9. `community_goals` & `community_goal_members` (`0006` , `0012` , `0014`)
- `community_goals` : `id UUID` , `name text` , `target bigint` , `unit text` , `ends_at timestampz` , `circle_id UUID` .
 - `community_goal_members` : Composite key `(goal_id, user_id)` . Tracks individual `progress bigint` .
 - RLS:** Viewable by all circle members; writes scoped to `auth.uid() = user_id` .
10. `focus_settings` & `focus_sessions` (`0007_focus.sql` , `0010_ibadah_lock.sql`)
- `focus_settings` : `duration_index int` .
 - `focus_sessions` : `goal_id UUID` references `adhkar_goals` , `duration_seconds int` , `completed boolean` , `started_at` , `ended_at` .
 - RLS:** Owner-only.
11. `prayer_calc_settings` (`0009_prayer_calc_settings.sql`)
- Primary key `user_id UUID` .
 - Columns: `latitude double precision` , `longitude double precision` , `city text` , `country text` , `timezone text` , `calculation_method text` , `madhab text` .
 - RLS:** Owner-only.

12. `blocked_apps` & `emergency_overrides` (`0010_ibadah_lock.sql`)
- `blocked_apps` : `user_id` UUID , `package_name` text , `app_name` text , `platform` text ('ios', 'android') .
 - `emergency_overrides` : `user_id` UUID , `session_id` UUID , `reason` text , `unlocked_at` timestampz .
 - **RLS**: Owner-only.
13. `prayer_alarm_settings` & `wake_verifications` (`0011_wake_alarm.sql` , `0016_two_stage_wake.sql`)
- `prayer_alarm_settings` : `user_id` UUID , `prayer_name` text , `enabled` boolean , `verification_token` text , `wudu_token` text .
 - `wake_verifications` : Append-only log of successful scans (`user_id` , `prayer_name` , `verified_date` , `verified_at` timestampz , `stage` text).
 - **RLS**: Owner-only. Append-only policy (`insert` allowed; `update` disallowed; duplicate scans ignore conflicts).
-

8. Custom Native Module Deep Dive (`modules/expo-ibadah-native`)

8.1. Android Native Implementation (Kotlin)

Registered via `modules/expo-ibadah-native/android` :

1. `IbadahBlockingAccessibilityService.kt` :
 - Manifest config: `android:accessibilityEventTypes="typeWindowStateChanged"` .
 - Package Queries Exemption: `<queries>` in `AndroidManifest` allows discovering launchable packages on Android 11+ (API 30+).
 - Reads blocked package list from `IbadahBlockingPrefs` (SharedPreferences).
 - When a blacklisted package moves to foreground, launches `BlockedActivity` .
 2. `BlockedActivity.kt` :
 - Full-screen programmatic UI in dark tones reminding user of their active prayer/focus session.
 - Embeds a Dhikr recitation challenge to unlock if urgently needed.
 3. `WakeAlarmScheduler.kt` & `WakeAlarmRingingService.kt` :
 - Invokes `AlarmManager.setAlarmClock()` passing a `PendingIntent` to `WakeAlarmReceiver` .
 - Receiver starts `WakeAlarmRingingService` as a Foreground Service with type `mediaPlayback` or `specialUse` .
 - Loops alarm audio and posts full-screen intent notification launching `WakeAlarmFullScreenActivity` .
 4. `WakeAlarmBootReceiver.kt` :
 - Listens for `android.intent.action.BOOT_COMPLETED` to reschedule alarms persisted in `WakeAlarmPrefs` .
-

8.2. iOS Native Implementation (Swift)

Registered via `modules/expo-ibadah-native/ios` :

1. `IbadahNativeModule.swift` :
 - Exports native methods: `isAppBlockingSupported()`, `pickAppsToBlock()`, `startBlocking()`, `stopBlocking()`.
2. `FamilyActivityPickerPresenter.swift` :
 - Hosts Apple's SwiftUI `FamilyActivityPicker` within a native UIViewController.
 - Lets the user select apps and categories without exposing private bundle identifiers.
3. `IbadahLockSupport.swift` :
 - Serializes and deserializes Apple's opaque `ApplicationToken` structures to base64 strings for storage in TypeScript/Postgres.
4. `ManagedSettingsStore` :
 - Applies `ManagedSettingsStore().shield.applications = selectedTokens`.
 - Apple's operating system natively shields the target applications with an OS-level lock screen.

8.3. TypeScript Bridge & Autolinking

In `modules/expo-ibadah-native/index.ts` :

```
export interface IbadahNativeContract {
  isAppBlockingSupported(): boolean;
  pickAppsToBlock(): Promise<SelectedApp[]>;
  startBlocking(packageNames: string[]): Promise<void>;
  stopBlocking(): Promise<void>;
  scheduleWakeAlarm(id: string, triggerMs: number, prayerName: string): Promise<void>;
  cancelWakeAlarm(id: string): Promise<void>;
  stopWakeAlarmRinging(id: string): Promise<void>;
  openExactAlarmSettings(): Promise<void>;
}
```

- Gracefully handles platform variations: functions return safe no-ops or fallbacks on unsupported platforms (e.g. web or standard simulator) without throwing runtime missing-symbol crashes.

9. Build, Deployment, Environment & Testing Procedures

9.1. Environment Variables Configuration

Create a `.env` file in the root directory based on `.env.example` :

```
# Supabase Configuration (Required)
EXPO_PUBLIC_SUPABASE_URL=https://your-project-ref.supabase.co
EXPO_PUBLIC_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

# Kalimat API Configuration (Required for Semantic Search)
EXPO_PUBLIC_KALIMAT_API_KEY=your_kalimat_api_key_here
```

Warning

Only keys prefixed with `EXPO_PUBLIC_` are bundled into the client build. **Never** include the Supabase `service_role` key in this repository.

9.2. Executing Database Migrations

Apply migrations to your live Supabase instance using the Supabase CLI:

```
# Link your local repository to your remote Supabase project
supabase login
supabase link --project-ref <your-project-ref>

# Push all migrations (0001 through 0016) in order
supabase db push
```

9.3. Building the Application

Because Ibtida contains custom Kotlin and Swift code (`modules/expo-ibadah-native`), it cannot run on **standard Expo Go**. You must use a Development Client or EAS Build.

Local Native Prebuild:

```
# Install dependencies
npm install

# Regenerate native /android and /ios folders
npx expo prebuild

# Run Android on connected device/emulator
npx expo run:android

# Run iOS on physical device (required for FamilyControls)
npx expo run:ios
```

EAS Cloud Build:

```
# Build Android Development Client APK
eas build --profile development --platform android

# Build iOS Development Client (Ad Hoc or Internal Distribution)
eas build --profile development --platform ios
```

9.4. Automated & Verification Test Suites

```
# 1. TypeScript Strict Static Typecheck
npm run typecheck      # tsc --noEmit (Must exit 0)

# 2. ESLint Code Standard Verification
npm run lint           # eslint . --ext .ts,.tsx (Must exit 0)

# 3. Jest Unit Test Suites
npm run test           # Runs service & prayer engine unit tests

# 4. Playwright End-to-End Test Suites
npm run test:e2e       # Comprehensive E2E flows
npm run test:smoke     # Critical smoke tests (@smoke tag)
```

10. Store Compliance, Limitations & Strategic Roadmap

10.1. App Store & Google Play Policy Compliance

1. Google Play Accessibility Service Policy:

- Android app blocking utilizes an `AccessibilityService`.
- **Mandatory Action:** The Google Play Store listing must include an explicit, prominent user disclosure video and privacy policy explanation declaring that the Accessibility API is used exclusively for self-imposed digital fasting and application blocking during prayer, not for disability assistance or user data collection.

2. Apple Family Controls Entitlement:

- iOS app blocking requires the `com.apple.developer.family-controls` entitlement.
- This entitlement is **not self-service**. The account owner must submit a request via developer.apple.com detailing the self-restriction/wellness use case. See [docs/ios-family-controls-entitlement.md](#) for full application templates.

3. Android Exact Alarm Permission:

- Android 12+ requires `SCHEDULE_EXACT_ALARM` or `USE_EXACT_ALARM`. Must be accompanied by clear in-app justification for religious prayer times.

10.2. Production Operational Roadmap

- ☐ **Native Android Alarm Wiring:** Complete the JS trigger in `wakeAlarmScheduling.ts` to execute `scheduleWakeAlarm` via the Kotlin native module, providing full-screen ringing alarm clock capabilities on Android.
- ☐ **Offline Synchronization Queue:** Introduce a persistent SQLite/WatermelonDB offline buffer to sync prayer logs and tasbeeh counts when reconnecting from dead zones.
- ☐ **Live Gyroscopic Qibla Compass:** Upgrade the static azimuth bearing card into an interactive real-time compass utilizing `expo-sensors` (magnetometer and accelerometer).
- ☐ **System-Aware Dark Mode Palette:** Connect `darkColors` from `src/theme/tokens.ts` to React Native's `useColorScheme()` across the remaining light-surface screens.

Conclusion & Handover Verification

This document represents the complete, unabridged architectural specification for **Ibtida**. All source files, database schemas, and native contracts documented herein are fully implemented, statically validated, and ready for deployment. For immediate inquiries or specialized module extensions, refer to the underlying inline documentation in [src/](#) and [modules/](#).